OUTPUT:

```
= SLIDING WINDOW PROTOCOL - FLOW CONTROL
  Enter total number of frames: 8
  Enter sender's window size : 4
  Enter frame number to simulate
  as lost
  [0 to enter - 1 for no loss) : 5

  Sender window: [Frame 0 - Frame 3]
  Node A ⟶ Sending frame 0
  Node A ⟶ Sending frame
  Node A ⟶ Sending frame 2
  Node A ⟶ Sending frame 3
  Node B ⟶ Received frame 0,
            Sending ACK 0
  Node B ⟶ Received Frame 1,
            Sending ACK 1
  Node B ⟶ Received frame 2,
            Sending ACK 2
```

Ex.No:7
[illegible]26

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

Aim:

To write a program to implement flow control data link layer using the slinding window.

Algorithm:

1) Start the program
2) Read the total number of frame to be sent (N), the sender's window size
3) Initialize base = 0 and nextseq = 0
4) Repeat while base < N
5) Repeat step 4 unit all N frames have been transmitted and acknowledged.
6) Stop the program.

Program:

```
def simulate (total_frame, lost frames
    base = 0
    next_seq = 0
    while base < total_frames
    window_end = min(base + window
                         _size-1)

    print ("\n.....")
    print (f" Sender Window: [frame ")
```

```
Sender window: [Frame 4 ~ Frame 7]
  Node A → Sending frame 5
  Node A → Sending frame 6
  Node A → Sending frame 7
  Node B → Received frame 4,
           Sending ACK 4
  Node B → Received frame 5,
           Sending ACK 5

  Node A →  ACK 7 received.
            Sliding window
              forward

All 8 frames successfully transmitted
from Node A to Node B.
```

```
print (f" (frame [next -{LOST In}"]
next_ seq += 1
If lost_frame is not None and box
   ⇐ lost_frame
print (f" Node B --> Frame{lost_frame}
print (f" Node A --> Time Out! Resending
next_seq = lost_frame
lost_frame = None
 Continue
 for f in range (base, next-seq):
    Print(f"Node B --> Received
                          Frame{f}
Sending ACK {f}")
    print(f " Node A }
    If _name_ == "_main_"
      main ()
```

Result:

The program to implement flow control at the data link layer using the Sliding window Protocol was successfully written in python executed and verified by simulating the flow of frames.